

Modeling of asphaltene deposition in oil reservoirs during primary oil recovery

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ABSTRACT

Asphaltene deposition in oil reservoirs during primary oil recovery and using the enhanced oil recovery methods are complicated and destructive problem with which most of the oil field in worlds has encountered. In this research, the examination of formation damage due to the asphaltene deposition, a four phase black oil model (oil, water, gas, and asphaltene) is developed in combination with proper thermodynamic and deposition models in cylindrical coordinates around a well in a reservoir. Subsequently, this model has been used for examining the effect of well production rate and the initial reservoir permeability on the asphaltene deposition behavior in a typical reservoir during primary production process. The results of model indicate that at a fixed permeability, with increase in the production rate, the amount of asphaltene deposits will increase. However, an increase in the reservoir permeability will decrease the deposition rate because of the existence of further channels for flow.

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1. Introduction

Crude oil is a mixture of various hydrocarbons including light to some heavier hydrocarbons and some heteromolecules including nitrogen, sulfur, and oxygen. The liquid state of oil is the results of the equilibrium of these components in the liquid and vapor phases. Adding of an outer agent like normal alkanes or hydrocarbon gases under some condition, leads to the disruption of this equilibrium and precipitation of asphaltene (Wang and Civan, 2001). Asphaltene can be defined as the fraction of crude oil that is insoluble in normal alkanes but soluble in toluene and benzene (Mitchell and Speight, 1973). The precipitation of asphaltenes can be ascribed to changes in crude oil composition caused by addition of lower boiling components as well as changes in temperature and pressure thereby the alteration of the complex equilibrium keeping the asphaltenes in solution or suspension. Phase behavior, viscosity and interface properties of crude oil are strongly influenced by asphaltene components.

Asphaltene precipitation is an important and serious problem with which the oil industry has encountered for a long time (Kawanaka et al., 1991). Asphaltene precipitation may occur in reservoirs during primary production and may deposit in the porous media of the oil reservoir (Papadimitriou et al., 2007). Asphaltene precipitation usually experiences in the reservoirs that the initial pressure is much larger than the bubble point pressure and the fluid in place is nearly light with small asphaltene content, during primary depletion (De Boer et al., 1995). Although, changing wettability may actually improve oil recovery, the deposited asphaltene on the formation rock

leads to the plugging of reservoir, alteration of rock surface wettability from water-wet to more oil-wet conditions and thereby lower the oil relative permeability (Nghiem et al., 1998; Pina et al., 2006). In petroleum literature, only limited amounts of experimental data reported on the asphaltene precipitation and deposition under reservoir conditions due to conducting laboratory tests on core samples is time-consuming and expensive. Carrying out field experiments for the recognition of this phenomenon is difficult and even impossible. Further developments of both the modeling and experimental measurements should be conducted to design comprehensive reservoir management strategies. The complexity of the process of asphaltene deposition and formation damage, make impossible to present a fully predictive model in the short term time frame, and the experimental data will need to be collected and then used to tune the models.

Since 1998 in which Leontaritis represented the first model for examining the effect of asphaltene deposition in oil reservoirs some researchers have attempted to model this phenomenon (Nghiem et al., 1998; Wang and Civan, 2001; Kohse and Nghiem, 2004; Almeida, 2004). Possessing the capability of predicting and simulating of the asphaltene deposition phenomenon, these researchers' models have some demerits. Some of these infirmities are the unstability of the model only after a few hours by Leontaritis (1998) and assuming the adsorption of asphaltene on the rocks as the dominant mechanism in the asphaltene deposition process by Nghiem et al. (1998). While according to visual experiment of Danesh et al. (1987), surface deposition, pore throat plugging and entrainment of deposits in asphaltene deposition process are more effective than the asphaltene adsorption.

Modeling of the asphaltene deposition in oil reservoirs still has room for further improvements. From a mathematical and numerical

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point of view, it is important to apply improved solution approaches that produce reliable results with reduced simulator runtime. This research attempts to combine the relevant thermodynamic and deposition models with a four phases, single well model in a cylindrical coordinate in order to study the behavior of asphaltene deposition during natural depletion of reservoir. A conventional finite difference, implicit pressure–explicit saturation (IMPES) formulation is used. The objective of developing such model is to obtain an improved black-oil formulation. The IMPES formulation is straightforward, requires less arithmetic per time step than other formulations, and has much smaller storage requirements than a fully implicit formulation. The main idea of the present study is to study the influence of the well production flow rate and formation permeability on the formation damage caused by asphaltene deposition. Furthermore, the effect of crude type on the model prediction in a typical reservoir is explored.

2. Theoretical calculation

In this study, the developed model is formed by combination of thermodynamic and deposition models in a black oil four phases simulator around a well in a reservoir. The various parts of this combinative model are:

2.1. Thermodynamic model for asphaltene precipitation

Study of crude oil phase behavior indicates that there are three factors in separating of solid asphaltene from oil, namely change in pressure, temperature and composition of oil components (Hirschberg et al., 1984; Burke et al., 1990; De Boer et al., 1995; Andersen and Speight, 1999; Thomas et al., 1992; Wang and Civan, 2001; Leontaritis, 1996; Paramanu et al., 1999). In natural depletion of reservoir, temperature is constant therefore the most important and effective factor in asphaltene precipitation is the reservoir pressure changing. In this research, according to the acceptance of reversibility of asphaltene precipitation (Danesh et al., 1987; Kokal et al., 1992; Ramos et al., 1997; Hammami et al., 2000; Ghiloum and Oskui, 2004; Beck et al., 2005), we use a thermodynamic model which presented by Hirschberg et al. (1984). Their model is based on Flory–Huggins theory assuming the asphaltene precipitation as a reversible process. They offered the following equation for determination of the volume fraction of dissolved asphaltene in oil (Hirschberg et al., 1984).

$$\phi_A = \left[\frac{V_A}{V_l} - 1 - \frac{V_A}{RT} (\delta_A - \delta_l)^2 \right] \quad (1)$$

where ϕ_A is the volume fraction of dissolved asphaltene in oil, V_A and V_l are the molar volume of asphaltene and oil respectively, δ_A and δ_l are the solubility parameters of asphaltene and oil respectively, T is absolute temperature and R is universal gas constant. This model has two adjustable parameters, namely V_A and δ_A which are determined by coincidence of the model results with the experimental data. The solubility parameter of liquid phase is defined by (Hildebrand, 1929):

$$\delta_L = \left(\frac{\Delta U_{\text{vaporization}}}{V_L} \right)^{1/2} \quad (2)$$

where $\Delta U_{\text{vaporization}}$ is the internal energy change during vaporization of a unit mole liquid. The approach of Jhaveri and Youngren (1988) in modifying the prediction of molar volume by PR EoS are accepted. In this approach a volume correction parameter is introduced in PR EoS. This parameter has the same unit as b_i in PR EoS and is defined as:

$$C_i = S_i b_i \quad (3)$$

where S_i is named shift parameter. The volume correction parameters are not changed the VLE conditions. The corrected liquid volume is determined by:

$$V_l^{\text{corr}} = V_l - \sum_i x_i C_i \quad (4)$$

The reported experimentally determined shift parameters for light hydrocarbons and the proposed equation for estimating shift parameter for heavier hydrocarbons are used in this study (Jhaveri and Youngren, 1988). The values of the binary interaction parameters for using in the EoS, k_{ij} , are determined as below:

$$k_{ij} = 1 - \left(\frac{2 \cdot v_{ci}^{1/6} \cdot v_{cj}^{1/6}}{v_{ci}^{1/3} + v_{cj}^{1/3}} \right) \quad (5)$$

This model is carried out by two sets of the experimental data represented by Burke et al. (1990) and Vafaie Sefti and Mousavi-Dehghani (2006). The critical volume of C_7^+ is determined by Riazi and Daubert (1987) correlations. Unfortunately, most crude oil characterization data provide only very little information on the heavy end components. Therefore, the acentric factor, and other critical properties of C_7^+ are adjusted to match the bubble pressure of crudes. In addition, for determining the model parameters for each sample, the accuracy of the model to predict the asphaltene phase behavior through comparison with the experimental data is examined. The verified model is used to simulate the asphaltene deposition process in an oil reservoir.

2.2. Deposition and plugging model

The presented models for examining the asphaltene deposition in cores are generally based on the models, which have been represented only for single phase fluid flow, because asphaltene is seen just in the oil phase. Most of these models are planned based on particle movement in porous media theory, which was presented by Gruesbeck and Collins (1982) (Almehaideb, 2004). Wang and Civan (2001) to examine of asphaltene deposition did some research and finally offered a new model for this phenomenon. With referring to three mechanisms presented before by Danesh et al. (1987) and according to the Gruesbeck and Collins theory, they could represent a nearly comprehensive model, which was developable for the whole of reservoir. Later, this model was used by some researchers such as Kohse and Nghiem (2004) and Almehaideb (2004) to model the asphaltene deposition in oil reservoirs (Kohse and Nghiem, 2004; Almehaideb, 2004). According to the previous researchers' results, Wang and Civan (2001) for driving the material balance equation assumed that the single phase dead oil is in the system. The core considered horizontally and therefore neglected the gravity forces. Considering Darcy law and material balance equations, they represented the following equation for asphaltene deposition rate:

$$\frac{\partial E_A}{\partial t} = \alpha C_A \varphi - \beta E_A (v_l - v_{cr,l}) + \gamma u_l C_A \quad (6)$$

where E_A is the volume fraction of deposited asphaltene on rock, C_A is the volume fraction of precipitated asphaltene, φ is the formation porosity, $v_l, v_{cr,l}$ are, respectively, the interstitial velocity of oil and its critical value; and u_l is Darcy velocity of oil.

In this equation, the first term represents the surface deposition. The second term represents the entrainment of asphaltene deposits by the flowing phase when the interstitial velocity is larger than a critical interstitial velocity. This term shows that the entrainment of asphaltene deposits is directly proportional to the amount of asphaltene deposits presented in the porous media, and the difference

Table 1
Composition and properties of crude oil samples.

Oil composition (mol%)	Oil 1 (Burke et al., 1990)	Oil 2 (Vafaei Sefti and Mousavi-Dehghani, 2006)
Nitrogen	0.57	0.27
Carbon dioxide	2.46	1.65
Hydrogen sulfide	–	0.03
Methane	36.37	30.24
Ethane	3.47	7.86
Propane	4.05	5.49
i-Butane	0.59	0.87
n-Butane	1.34	2.5
i-Pentane	0.74	1.03
n-Pentane	0.83	1.49
Hexane	1.62	7.38
Heptane plus	47.96	41.19
Total	100	100
<i>Oil properties</i>		
Heptane plus molecular weight	329	350
Heptane plus specific gravity	0.9594	0.9687
Live oil molecular weight	171.4	165
API gravity of stock tank	19	23
Asphaltene content in stock tank condition (wt.%)	16.08	14.3
Reservoir temperature, °F	212	205
Saturation pressure (reported)	2950	2100
Saturation pressure (estimated)	2050	2100

between the actual interstitial velocity and the critical interstitial velocity is necessary for asphaltene deposit mobilization. The last term represents the pore throat plugging rate, which is directly proportional to the superficial Darcy velocity and the concentration of solid asphaltene (Wang and Civan, 2001).

The second and the last terms of Eq. (6) are the correction of Gruesbeck and Collins model (Wang and Civan, 2001). This model has three parameters: α is the surface deposition rate constant, β is the entrainment of deposits constant and γ is the pore throat plugging rate constant. Wang and Civan (2001) were offered relations for description of β and γ .

The plugging model refers to the effect of asphaltene deposition on the porosity and permeability (Almeida, 2004). Various investigators have indicated the porosity is reduced by the amount of the fractional pore volume occupied by asphaltene deposit.

$$\varphi = \varphi_0 - E_A \quad (7)$$

where φ_0 is the initial porosity.

The effect of deposition on the permeability is modeled in variety of ways (Ali and Islam, 1997; Nghiem et al., 1998; Minssieux et al., 1998). Civan et al. (1989) using the following equation for represent-

ing the behavior of permeability drop as a porosity function (Civan et al., 1989):

$$k = k_0 \left(\frac{\varphi}{\varphi_0} \right)^3 \quad (8)$$

where k_0 is the initial permeability.

Chang and Civan (1997) emended Eq. (8) to account the pore throat plugging as follows:

$$k = f_p k_0 \left(\frac{\varphi}{\varphi_0} \right)^3 \quad (9)$$

where f_p is the permeability correction constant.

They commented that this equation could be used for showing the effect of pore throat plugging in various real porous media. The relative permeability will play a significant role in the make-up of the produced fluid. However, if asphaltenes are to adsorb onto surfaces, the wettability and the relative permeability may also be affected. This may induce significant plugging of the oil phase, and blocks all mobile phase approximately equally (Nghiem et al., 1998). Another permeability correction constant f_w is introduced in Eq. (9) to account the effect of wettability alteration on the permeability drop as follow:

$$k = f_p f_w k_0 \left(\frac{\varphi}{\varphi_0} \right)^3 \quad (10)$$

More studies are required for estimation of the permeability correction constants as well as the deposition and plugging model parameters. These adjustable parameters are essentially obtained by matching the experimental and numerical results. However, the effect of wettability alteration on the permeability drop is not considered in this paper.

2.3. Single well model

Thereto precipitation and deposition models that descript before, to show multiphase flow through porous media we need to use the mass conservation and Darcy law (Almeida, 1989). The simulating fluid flow models through porous media set to black oil models and compositional models. Almeida (2004) presented a four phase single well model to simulate asphaltene deposition in the near wellbore region. He extended a single well model and assumed that asphaltene to be the fourth component and treating it in a similar fashion to the treatment of free and dissolved gas (Almeida, 2004). In this research, the mass conservation equations for the four phases flow through the porous media with the Almeida (2004) assumptions relative to the presence of asphaltene are used to model the asphaltene deposition in petroleum reservoirs. For four phases (oil, water, gas, and asphaltene) black oil model assuming the asphaltene and gas are both free in the reservoir and dissolved in the oil phase the flow equations are represented as follows:

Oil:

$$\frac{1}{r} \left[r \lambda_o \left(\frac{\partial p_o}{\partial r} - \gamma_o \frac{\partial D}{\partial r} \right) \right] + \frac{\partial}{\partial z} \left[\lambda_o \left(\frac{\partial p_o}{\partial z} - \gamma_o \frac{\partial D}{\partial z} \right) \right] - q_o = \frac{\partial}{\partial t} \left(\frac{\varphi S_o}{B_o} \right) \quad (11)$$

Water:

$$\frac{1}{r} \left[r \lambda_w \left(\frac{\partial p_w}{\partial r} - \gamma_w \frac{\partial D}{\partial r} \right) \right] + \frac{\partial}{\partial z} \left[\lambda_w \left(\frac{\partial p_w}{\partial z} - \gamma_w \frac{\partial D}{\partial z} \right) \right] - q_w = \frac{\partial}{\partial t} \left(\frac{\varphi S_w}{B_w} \right) \quad (12)$$

Table 2
Thermodynamic model parameters.

Parameter	Oil 1	Oil 2
δ_A (Mpa ^{0.5})	21.786	22.53
V_A (m ³ /kmol)	1.556	1.6564

Gas:

$$\begin{aligned} & \frac{1}{r} \left[r \lambda_g \left(\frac{\partial p_g}{\partial r} - \gamma_g \frac{\partial D}{\partial r} \right) + r R_s \lambda_o \left(\frac{\partial p_o}{\partial r} - \gamma_o \frac{\partial D}{\partial r} \right) \right] \\ & + \frac{\partial}{\partial z} \left[\lambda_g \left(\frac{\partial p_g}{\partial z} - \gamma_g \frac{\partial D}{\partial z} \right) + \lambda_o R_s \left(\frac{\partial p_o}{\partial z} - \gamma_o \frac{\partial D}{\partial z} \right) \right] \\ & - q_g - R_s q_o = \frac{\partial}{\partial t} \left(\frac{\phi S_g}{B_g} + \frac{R_s \phi S_o}{B_o} \right) \end{aligned} \quad (13)$$

Asphaltene:

$$\begin{aligned} & \frac{1}{r} \left[r (\lambda_A + R_A \lambda_o) \left(\frac{\partial p_o}{\partial r} - \gamma_o \frac{\partial D}{\partial r} \right) \right] \\ & + \frac{\partial}{\partial z} \left[r (\lambda_A + R_A \lambda_o) \left(\frac{\partial p_o}{\partial z} - \gamma_o \frac{\partial D}{\partial z} \right) \right] \\ & - \left(\frac{S_A}{S_o} + R_A \right) q_o = \frac{\partial}{\partial t} \left(\frac{\phi S_A}{B_A} + \frac{R_A \phi S_o}{B_o} + \frac{E_A}{B_A} \right) \end{aligned} \quad (14)$$

where P , S , q , and B are, respectively, the pressure, saturation, production rate, formation volume factor of each of the fluid phases; D is the depth (positive downward); r and z are the radial and vertical dimensions; S_A indicates the suspended asphaltene saturation and R_A is the volume ratio of soluble asphaltene in the oil phase.

The saturation equations for the flowing phases are related through:

$$S_o + S_w + S_g + S_A = 1.0 \quad (15)$$

The phase pressures in Eqs. (11)–(14) are related through the capillary pressures, which are functions of saturations as:

$$P_{cow} = P_o - P_w \quad (16)$$

$$P_{cgo} = P_g - P_o \quad (17)$$

The mobility of each phase is calculated based on the following equation:

$$\lambda_i = \frac{k k_{rl}}{B_i \mu_i} \quad (18)$$

where λ is the phase mobility and i denotes each of the fluid phases.

If we assume that the velocity of oil and precipitated asphaltene are the same (because asphaltene is presence in the oil phase and flow

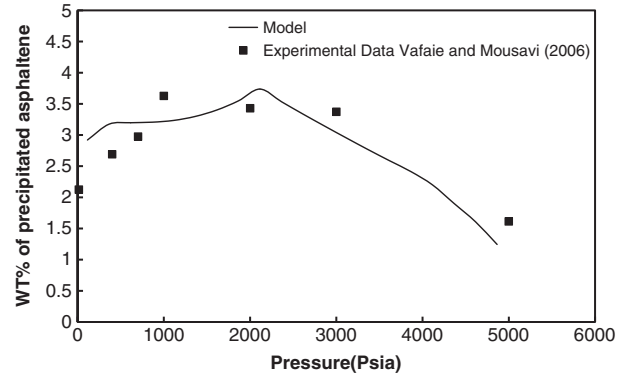


Fig. 2. Pressure effect on the amount of asphaltene precipitation in oil 2 at 205 °F.

with that), we can represent a new definition for the mobility of oil and asphaltene as follows:

$$\lambda_o = \frac{k k_{ro}}{\mu_o} \left(\frac{S_o}{S_o + S_A} \right) \quad (19)$$

$$\lambda_A = \frac{k k_{ro}}{\mu_o} \left(\frac{S_A}{S_o + S_A} \right) \quad (20)$$

where k , k_r and μ are, respectively, the absolute permeability, relative permeability, and viscosity of the denoted phases.

In addition, the gravity term is determined by following equation:

$$\gamma_l = \rho_{l, std} \times \frac{g}{g_c} \sin(\theta) \quad (21)$$

where $\rho_{l, std}$ is the phase density under standard condition; and θ is the deviation angle.

For calculating the amount of asphaltene deposited in porous media (in Eq. (14)) we can use this equation (Almehaideb, 2004):

$$E_A = E_A + \left(\frac{\partial E_A}{\partial T} \right) \Delta T \quad (22)$$

2.3.1. Initial and boundary conditions

The proper initial and boundary conditions are need for calculating system of Eqs. (1)–(22) for a well located in a reservoir. In this research, we use the following initial and boundary conditions.

Because at the start point of production usually reservoir pressure is above the upper asphaltene precipitation pressure then, at $t = 0$, the volume fraction of suspended asphaltene in the whole of the reservoir is zero.

$$C_A = 0, 0 < r < r_e, 0 < z < h, t = 0 \quad (23)$$

Table 3

Basic data for the reservoir in all the runs.

Reservoir characteristic	
r_w (ft)	0.5
r_e (ft)	10,000
Reservoir height (ft)	50
Initial reservoir pressure (psia)	5260
Porosity (%)	25
S_{or}	0.2
S_{wirr}	0.2

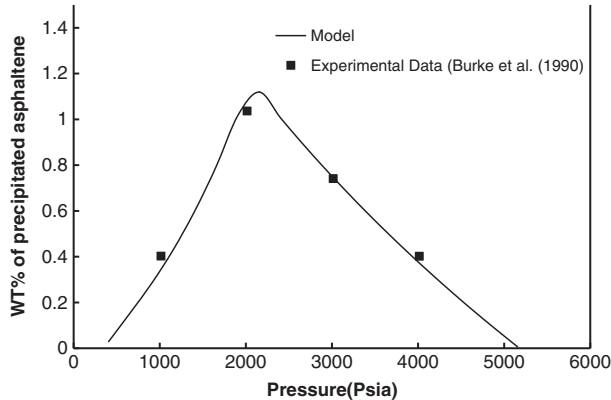


Fig. 1. Pressure effect on the amount of asphaltene precipitation in oil 1 at 212 °F.

Table 4
Deposition model constants.

Parameter	Value
$\alpha(1/\text{day})$	0.01
$\beta(1/\text{ft})$	0
$\gamma_i(1/\text{ft})$	0
σ, cte	0
$f_{p, \text{cte}}$	1
$f_{M, \text{cte}}$	1

If there is no precipitated asphaltene, the volume fraction of asphaltene deposited related to Eq. (6) is zero; therefore, there is no reduction in the permeability and porosity values.

$$E_A = 0, 0 < r < r_e, 0 < z < h, t = 0 \quad (24)$$

$$\phi = \phi_0, 0 < r < r_e, 0 < z < h, t = 0 \quad (25)$$

$$k = k_0, 0 < r < r_e, 0 < z < h, t = 0 \quad (26)$$

where r_e is the drainage radius of well; and h is the height of oil zone.

The inner boundary condition is defined by constant total oil flow rate for the well. Based on this condition, the phase mobility is determined as follow:

$$T_{\text{Re}} = \frac{n \Delta z C_1}{\ln\left(\frac{r_1}{r_w}\right)} \quad (27)$$

$$T_o^n = T_{\text{Re}}^n \times \lambda_o^n \quad (28)$$

where T_{Re} is the geometric factor for the layer n , C_1 is a conversion factor for units other than S.I, T_o is the oil transmissibility and r_w is the well radius.

Therefore, the oil flow rate at each layer for a well block is calculated as follow:

$$Q_o^n = \frac{T_o^n}{\sum_n T_o^n} Q_{o, \text{tot}} \quad (29)$$

where $Q_{o, \text{tot}}$ is the total oil flow rate for all layer of the reservoir.

Having the analogous definition with T_o for other three phases, we can calculate the production rate of each phase in each layer.

$$Q_g^n = \frac{T_g^n}{T_o^n} Q_o^n + R_s^n Q_o^n \quad (30)$$

$$Q_w^n = \frac{T_w^n}{T_o^n} Q_o^n \quad (31)$$

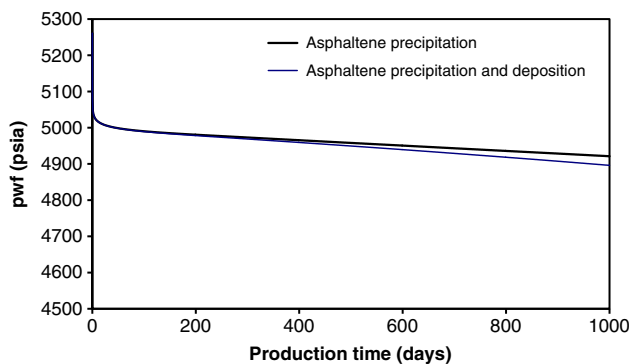


Fig. 3. Flowing bottomhole pressure versus production time for oil 1, with initial reservoir permeability 25 md, and well flow rate 500 STB/day.

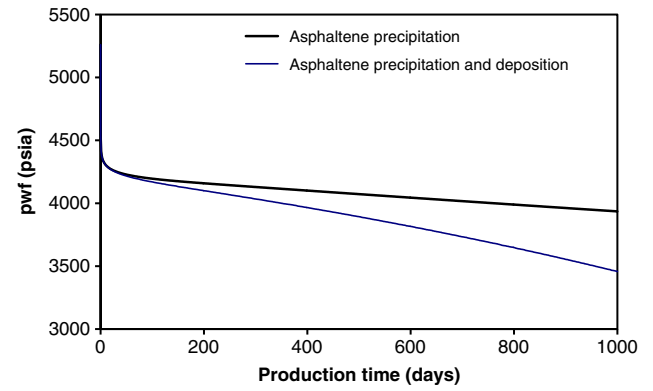


Fig. 4. Flowing bottomhole pressure versus production time for oil 1, with initial reservoir permeability 25 md, and well flow rate 2000 STB/day.

$$Q_A^n = \frac{T_A^n}{T_o^n} Q_o^n + R_A^n Q_o^n \quad (32)$$

where R_s^n is the solution gas oil ratio in each layer. The outer boundary condition is a no-flow boundary.

3. Solution method

Eqs. (11)–(32) with the thermodynamic model (Eq. (1)) and the deposition and plugging models (Eqs. (6), (7) and (9)) form black oil four phases simulator around a productive well are solved simultaneously for showing the behavior of asphaltene deposition in reservoir and studying the affective parameters on this phenomenon. IMPES method is selected to solve this system of coupled nonlinear equations (Fanchi et al., 1982). In this method, first pressure (oil pressure) distribution in the whole of reservoir is determined implicitly. To determine the oil pressure, Eqs. (11)–(14) are recombined to obtain one linear equation with only one unknown for each grid block that is the oil pressure. With having all necessary variables in the previous time step and solving the pressure equation, the pressure distributions around a well are calculated.

Then, the saturation distributions for all four phases are determined explicitly. With having pressure in all block, thermodynamic model calculates the volume fraction of asphaltene dissolved in the oil and precipitated from oil. Then, the deposition model calculates the amount of deposited asphaltene on the rock. Porosity and permeability distribution are then determined by the plugging model. Finally,

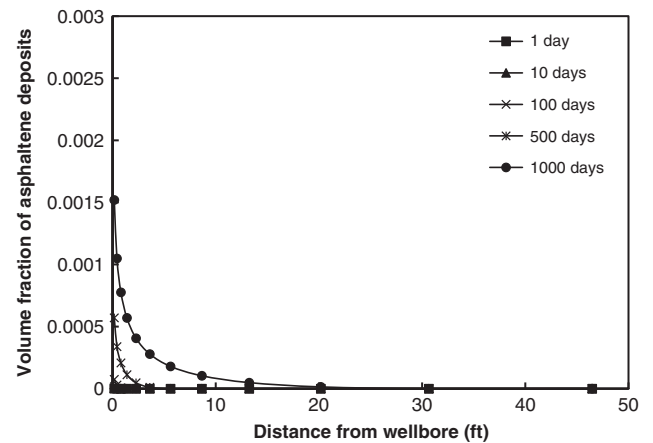


Fig. 5. Asphaltene deposits around a well during production for 500 STB/day and initial reservoir permeability 25 md.

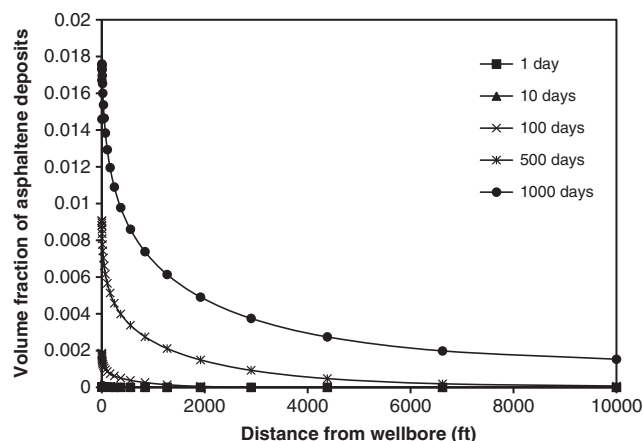


Fig. 6. Asphaltene deposits around a well during production for 5000 STB/day and initial reservoir permeability 25 md.

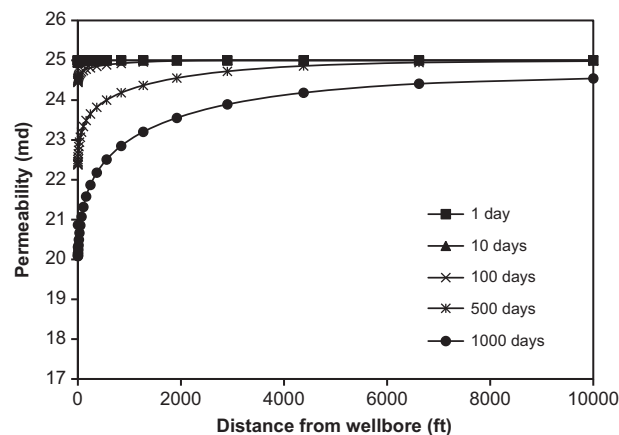


Fig. 8. Permeability distribution after deposition around a well at 5000 STB/day production.

other model parameters are calculated and these values of the model parameters are used in the new time step with repetition of the computations.

4. Case studies

Two sets experimental data of asphaltene precipitation at different pressure presented by Burke et al. (1990) and Vafaie Sefti and Mousavi-Dehghani (2006) are used to verify the accuracy of the asphaltene thermodynamic model. These results illustrate the primary depletion process of a reservoir. Burke et al. (1990) and Vafaie Sefti and Mousavi-Dehghani (2006) oils are employed in this combinative model to show the formation damage calculation features due to the asphaltene deposition in a typical reservoir. Composition and properties of these samples are listed in Table 1. Fractions that to be higher-boiling than heptane are grouped in one pseudo component. The determined molar value and solubility parameters of both crudes asphaltene by matching the experimental data are listed in Table 2. The formation volume factor of asphaltene is set equal to 1 due to it is assumed that asphaltene is an incompressible phase. The solution gas–asphaltene-ratio is assumed to be equal zero, as we know, compositionally, that heavy hydrocarbons components present in asphaltenes. It can be assumed that the amount of gas in asphaltene is negligible.

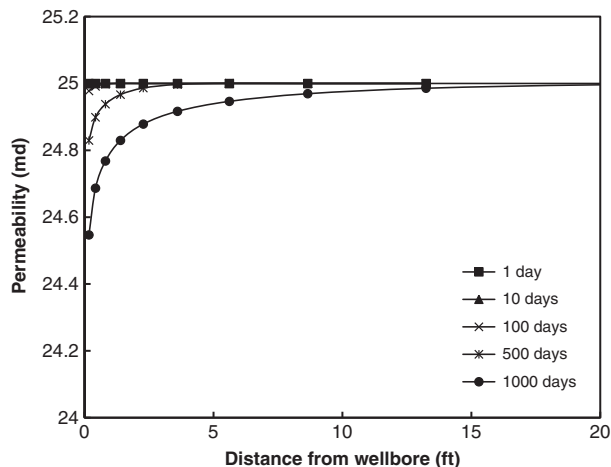


Fig. 7. Permeability distribution after deposition around a well at 500 STB/day production.

There is no more experimental data for asphaltene deposition and so no suitable values for model parameters like that the interstitial velocity and critical pore throat diameter are known. Therefore, we assume that the surface deposition as dominant mechanism, which was previously proposed (Wang and Civan, 2001). The surface deposition constant is set equal to 0.01 1/day in all the model runs.

5. Result and discussion

5.1. Thermodynamic model predictions

The significant parameter that affects the asphaltene precipitation is pressure (Hirschberg et al., 1984; Burke et al., 1990) especially during primary depletion process. During the production process, the reservoir pressure is a factor that has more changing. Many investigators studied the effect of pressure on the asphaltene phase behavior. They found that bubble point pressure is an important factor in asphaltene precipitation process. Because at this pressure the dissolved gas in oil begins to separate from oil and therefore, with decrease the pressure below the bubble point, the oil phase become heavy and so the difference between the oil solubility parameter and asphaltene solubility parameter will be small. Referred to Eq. (1), below the bubble point pressure, with decrease in the reservoir pressure the amount of precipitated asphaltene will decrease. Above the bubble point pressure, with decrease in the pressure molar volume of oil will increase and the asphaltene solubility will decrease.

Figs. 1 and 2 show the effect of pressure on the amount of asphaltene precipitation for oil 1 at 212 °F and oil 2 at 205 °F

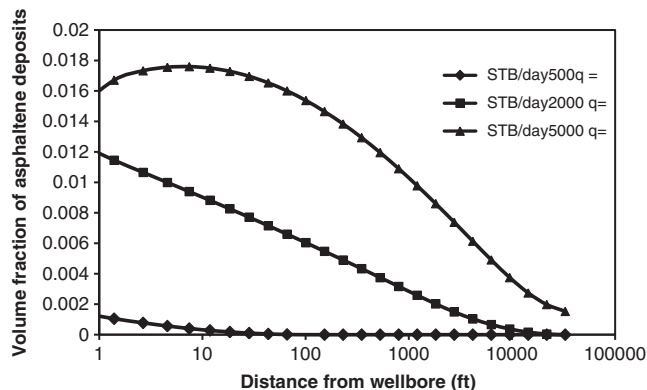


Fig. 9. Asphaltene deposits profile around a well with production flow rate after 1000 days with initial reservoir permeability 25 md.

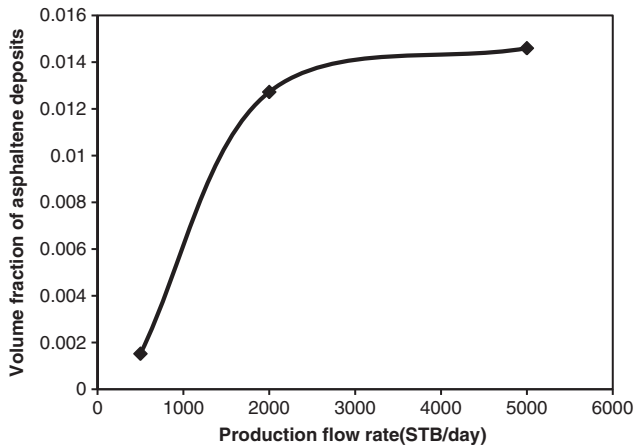


Fig. 10. Asphaltene deposits with production flow rate at wellbore after 1000 days with initial reservoir permeability 25 md.

respectively. As demonstrated in these two figures, a reasonable matching of the experimental data and our model has existed.

5.2. Reservoir model

In order to illustrate the application of the present model for studying the effects of asphaltene deposition on the formation damage qualitatively, the model is run during primary recovery process under different initial reservoir permeability condition and various well flow rates. The reservoir characteristic parameters and the deposition model constants, which are used in the model, are listed in Table 3 and 4 respectively.

5.2.1. Effect of asphaltene deposition on flowing bottomhole pressure

An attempt is made to determine the effect of involving the deposition and plugging model in the main proposed model on the flowing bottomhole pressure (Pwf). Figs. 3 and 4 show the production time dependence of the flowing bottomhole pressure predicted for the cases after asphaltene precipitation, the deposition of solid in reservoir occurred or not respectively at initial reservoir permeability 25 md, well flow rate 500 and 2000 STB/day.

As it is observed in Fig. 3 when there is not deposited asphaltene during the production time, the trend of Pwf decreases rapidly and reaches a steady state value after a few days, which depends to the well flow rate and the reservoir permeability. When the reservoir is

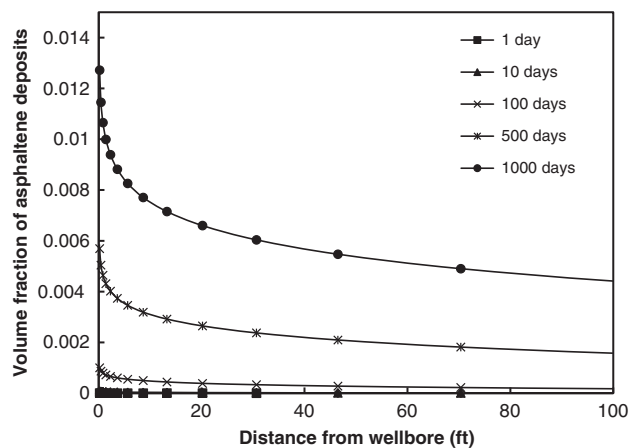


Fig. 11. Asphaltene deposits in reservoir at different production time with initial reservoir permeability 25 md and 2000 STB/day production.

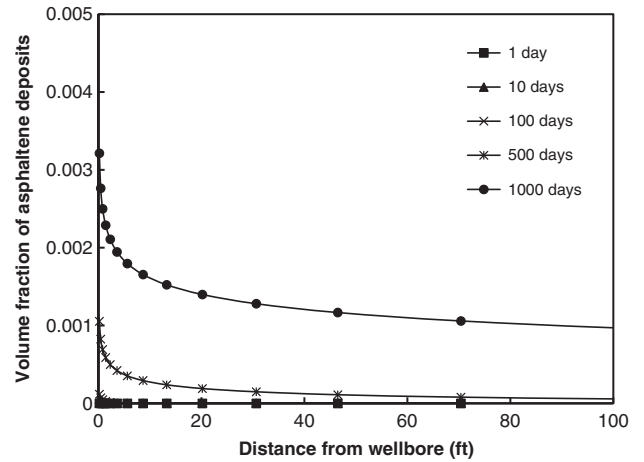


Fig. 12. Asphaltene deposits in reservoir at different production time with initial reservoir permeability 25 md and 2000 STB/day production.

encountered with the asphaltene deposition phenomenon, the deposited asphaltene will cause an extra and continuous decrease in Pwf through the obstructing the pores compared to the conditions in which deposition is not taken place. A similar behavior is observed in Fig. 4, where a higher well flow rate provides faster flowing bottomhole pressure decreasing rate.

5.2.2. Effect of well flow rate on deposition behavior

The deposition of asphaltene in the near wellbore region after various production times at initial reservoir permeability 25 md and for 500 and 5000 STB/day gives results as shown in Figs. 5 and 6 respectively.

Ali and Islam (1997) showed the effect of flow rate on the asphaltene deposition behavior. In this study, for surveying the well flow rate on the asphaltene deposition, the model is run for two well flow rates at a fixed permeability. Results are qualitatively in good agreement with Ali and Islam (1997). It is observed that with an increase in the well flow rate, because of further pressure loss and therefore further solid asphaltene precipitation, the accumulated deposit will increase and the porosity and permeability of the reservoir are damaged.

Figs. 5 and 6 show the effect of well flow rate on the cumulative asphaltene deposits around a well during production. As can be followed in Figs. 5 and 6 with increased well flow rate due to further pressure loss, the solid asphaltene concentration and the cumulative asphaltene deposit are increased at each production time around the well. In addition, at low flow rates the boundary of deposits is less than the higher flow rates. This means that with an increase in the well flow rate, the boundary of deposits are propagated and can inclusive of the entire reservoir.

Figs. 7 and 8 show the permeability distribution of different production rate after deposition phenomenon. As seen in these figures, with an increase in the production rate, the permeability loss and the boundary of deposits increase. In addition, at low production rate, maximum permeability loss is about 0.5 md but for high production rate, permeability loss is about 5 md. This is due to the dependence of the permeability on the deposits of asphaltene (Eqs. (3) and (5)).

Our model provides a reasonable explanation for the effects of production flow rate and pressure depletion on the asphaltene deposition behavior in reservoir. Fig. 9 indicates the result of well flow rate on deposition at initial reservoir permeability 25 md after 1000 days respectively. The predicted results in Fig. 9 demonstrate that the amount of deposited asphaltene due to the pressure depletion increases with increased well flow rate. There is a maximum value of deposited asphaltene in Fig. 9 around a radius 1.4 ft at

Table 5
Effects of the initial reservoir permeability and production flow rate on the permeability and porosity reduction percent in the near wellbore region for two crude oil samples.

Oil	Initial permeability, md	25			100			1000		
	Flow rate, STB/day	500	2000	5000	500	2000	10,000	5000	10,000	50,000
No. 1	Permeability reduction %	0.5907	10.7	18.84	0	2.51	16.27	3.18	6.61	16.64
	Porosity reduction %	0.2	3.71	6.72	0	0.8	5.75	1.07	2.22	5.88
No. 2	Permeability reduction %	2.7	12.67	22.53	0	6.68	16.7	7.12	9.47	18.86
	Porosity reduction %	0.91	4.41	8.17	0		5.95	2.43	3.26	6.73

5000 STB/day. The pressure at this radius is computed as the bubble pressure of reservoir fluid. This means that the porosity and permeability at this radius decreased severely. This tendency is consistent with the precipitation thermodynamic behavior of crude presented in Fig. 1. As demonstrated in Fig. 10, it is expected that with an increase in the well flow rate, the volume fraction of deposited asphaltene for oil 1 reaches a plateau off after 1000 days at wellbore radius with 25 md initial permeability.

5.2.3. Effect of initial reservoir permeability on deposition behavior

It is clear that a larger initial reservoir permeability value represents a reservoir with more channels for flow of fluids and consequently lesser pressure drop. Figs. 11 and 12 show the amount of asphaltene deposit in relation to distance from the wellbore at 2000 STB/day production and different production time for 25 and 100 md respectively. As the production time increases, the amount of asphaltene deposit increases. Comparison of these figures reveals that, significantly lesser-accumulated deposits around a well might be expected with higher value of initial reservoir permeability than one with lower value in the all production time. This observation can be attributed to the lesser asphaltene precipitation in higher permeable reservoirs. The permeability reduction percent as a criterion of the formation damage due to the asphaltene deposition is computed 4% for the initial permeability reservoir 100 md, while this value is 14% for 25 md. In fact, with increase in the initial reservoir permeability formation damage due to the asphaltene deposition will decrease. Other author found the same behavior for a similar system (Minssieux et al., 1998).

5.2.4. Effect of crude oil nature on deposition behavior

The effect of crude oil nature on wellbore damage is studied using the phase behavior of two crude samples. The thermodynamic model predicts that the precipitated solid for the oil 2 is greater than the oil 1 at the same temperature and pressure. Therefore, it is supposed to predict the deposition behavior as the phase behavior while the asphaltene content of oil 2 is lesser than the oil 1. The predicted results of the deposition behavior of oils are compared in Table 5. The results show that the amount of asphaltene deposit from oil 2 to be greater than the oil 1 at different radius around wellbore under the same reservoir characteristic and production flow rate conditions.

The effects of initial reservoir permeability and production flow rate on the permeability and porosity reduction percent in the near wellbore region for two crude oil samples are presented in this table. The permeability and porosity reduction percent of the reservoir under flowing condition of oil 2 are greater than the oil 1. Crude that is more unstable causes more reservoir damage by asphaltene deposition. At the initial permeability of 100 md and production rate of 500 STB/day the amount of deposit is nearly zero. Therefore, at an initial permeability of the reservoir there is a minimum production flow rate with no deposition problem. This is a good remark for controlling the asphaltene deposition behavior in the reservoir under the computed condition.

6. Conclusions

In this paper, the proposed model was used on two examples cases to study the effects of production rate and reservoir permeability on the asphaltene deposition in flow restrictions. Making use the Flory–Huggins model enables us to simulate the empirical data's for asphaltene precipitation truly. The effect of asphaltene deposition on the behavior of bottomhole pressure indicates the effect of this phenomenon on formation damage because; in a reservoir with asphaltene deposition phenomenon, there is an extra pressure loss. The initial permeability of reservoir has an important role on the behavior of asphaltene deposition. With an increase in the initial permeability at a fixed production rate, formation damage will be considerably less, because of the availability of more channels for flow and decrease in the pressure. Most important factor in the asphaltene deposition process is the well production rate. With more production rate at the fixed initial reservoir permeability, there is more difficulties in asphaltene reservoirs. If in a reservoir that has asphaltene deposition risk, the production rate will align with the reservoir permeability (formation capacity), there is minor difficulties due to the asphaltene deposition.

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